Assignment 6: Connect and Test a Push Button with Arduino

Internship: IoT with AI

Week: 1

Topic: Introduction to IoT and Electronics Basics

1. Introduction

A push button is a simple electronic input component used to control electronic circuits. It allows users to provide input to a microcontroller by pressing or releasing the button.

In Arduino-based projects, push buttons are commonly used to control LEDs, motors and other electronic devices.

2. Objective

To connect a push button with Arduino Uno and understand how digital input signals can be used to control an LED.

3. Components Required

1. Arduino Uno R3 – 1
2. Push button – 1
3. LED – 1
4. 220Ω resistor – 1
5. Breadboard – 1
6. Jumper wires – As required
7. Tinkercad Circuits – Online simulation platform

4. Circuit Connections

The push button and LED are connected to Arduino Uno as follows:

Push Button Connections:

- One terminal of the push button is connected to digital pin 2.
- The opposite terminal is connected to GND.
- Arduino internal pull-up resistance is enabled using INPUT_PULLUP.

LED Connections:

- The LED anode (long leg) is connected to digital pin 13 through a 220Ω resistor.
- The LED cathode (short leg) is connected to GND.

5. Working Principle

The Arduino reads the push button status using the digitalRead() function.

The INPUT_PULLUP mode enables an internal pull-up resistor in Arduino.

- When the button is released, the input reads HIGH.
- When the button is pressed, the input reads LOW.
- When the button is pressed, Arduino switches ON the LED.
- When the button is released, Arduino switches OFF the LED.

6. Arduino Program

```
int buttonPin = 2;
int ledPin = 13;

void setup() {
  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(ledPin, OUTPUT);
```

```
}

void loop() {
  int buttonState = digitalRead(buttonPin);

  if (buttonState == LOW) {
    digitalWrite(ledPin, HIGH);
  }
  else {
    digitalWrite(ledPin, LOW);
  }
}
```

7. Simulation Procedure

1. Open Tinkercad Circuits.
2. Create a new circuit and add Arduino Uno.
3. Place a push button on the breadboard across the centre gap.
4. Connect the push button to digital pin 2 and GND.
5. Place an LED and connect it to digital pin 13 through a 220Ω resistor.
6. Connect the LED cathode to GND.
7. Open the Code editor and select Text mode.
8. Enter the Arduino program.
9. Start the simulation.
10. Press and release the push button to observe the LED response.

8. Expected Output

- Button pressed: LED ON.
- Button released: LED OFF.

The LED responds immediately to the push button input.

9. Applications

1. Doorbell systems.
2. Smart home automation.
3. Electronic control panels.
4. Digital switching systems.
5. Embedded system applications.

10. Conclusion

This experiment demonstrates how a push button can be interfaced with Arduino Uno as a digital input device. The Arduino reads the button status and controls the LED accordingly. The experiment provides practical understanding of digital input, internal pull-up resistors and digital output programming.